Automatic Classification of Palatal and Pharyngeal Wall Shape Categories from Speech Acoustics and Inverted Articulatory Signals

Ming Li[123], Adam Lammert[1], Jangwon Kim[1]
Prasanta Ghosh[4], Shrikanth Narayanan[1]

1 Signal Analysis and Interpretation Laboratory, University of Southern California, Los Angeles, USA
2 Sun Yat-Sen University Carnegie Mellon University Joint Institute of Engineering
Sun Yat-Sen University, GuangZhou, China
3 Sun Yat-Sen University Carnegie Mellon University Joint Research Institute, ShunDe, China
4 Department of Electrical Engineering, Indian Institute of Science (IISc), Bangalore, India

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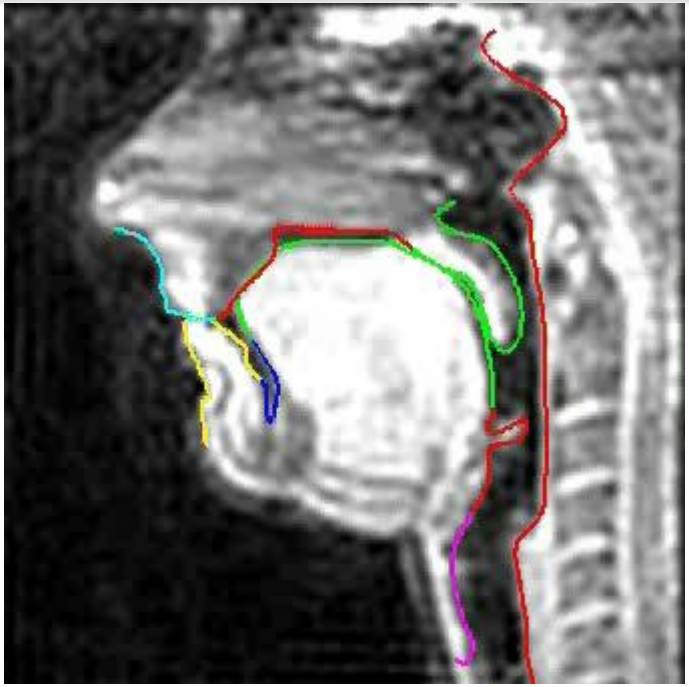
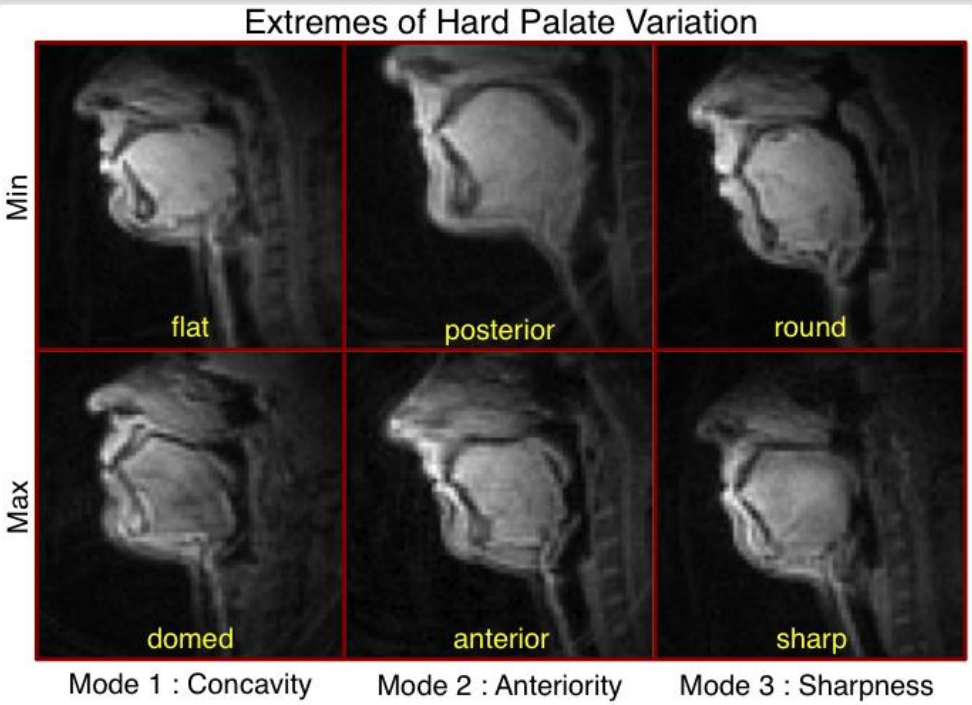
Morphological variability





- Vocal tract morphological variability
  - Vocal tract length (Peterson, 1952; Fant, 1960, Lee, 1999, Stevens, 1998)
  - Shapes of hard palate and posterior pharyngeal wall (Lammert, 2011)
  - Normalization for ASR (VTLN, etc.) or features for Speaker ID

Articulatory variability

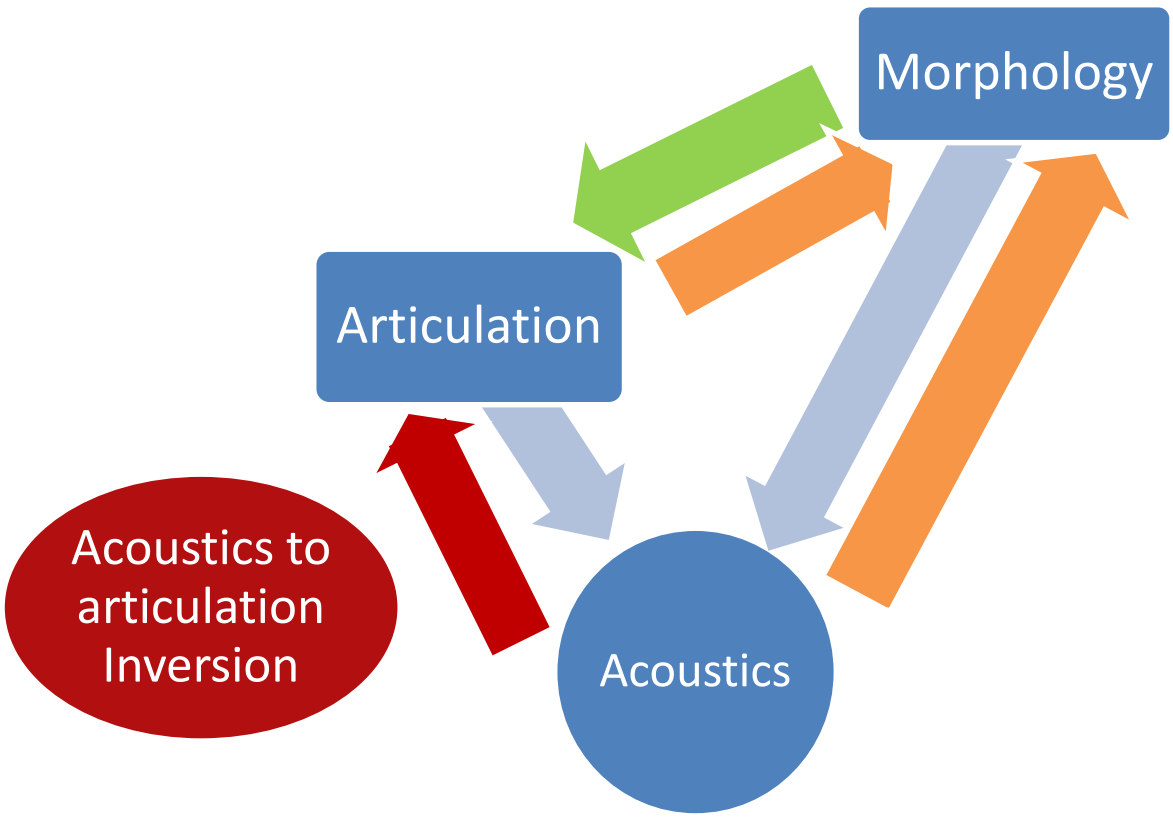
- Morphological variability also reflects on articulation
  - Flat palates exhibit less articulatory variability than highly domed palates during vowel production (Perkell, 1997; Mooshammer, 2004; Brunner, 2005; Brunner, 2009)
  - Articulation of coronal fricatives is influenced by palate shape
    - apical vs. laminal articulation of sibilants (Dart, 1991)
    - jaw height and tongue body positioning (Honda, 2002, Thibeault, 2011).
  - An example in Singing

We believe that in order to make standard pronunciation, articulation need to compensate the vocal tract morphology.

- Different speakers articulate differently even for the same words

Morphology, articulation, acoustics

- Acoustics + inverted articulation -> morphology



Methods, database and results

- Methods
  - Open-smile features + GMM UWPP supervectors with SVM
  - Feature level fusion of MFCC + inverted articulatory features
- Database
  - In-house collected real time MRI data (rtMRI)
    - With accurate morphology measurement
    - Noisy speech
  - Wisconsin X-Ray Microbeam data (XRMB)
    - With morphology measurement (sensor placement may vary)
    - Clean speech
- Experimental results
  - 63% ~ 71% accuracy for concavity binary classification
  - Appending inverted articulatory features with MFCC 66%->69%

Thank you for your attention

- Any questions or comments?
- For more details, please refer to our poster and paper.